



This visual does not support exporting.

500K

Total Accidents

2.50

Avg Severity

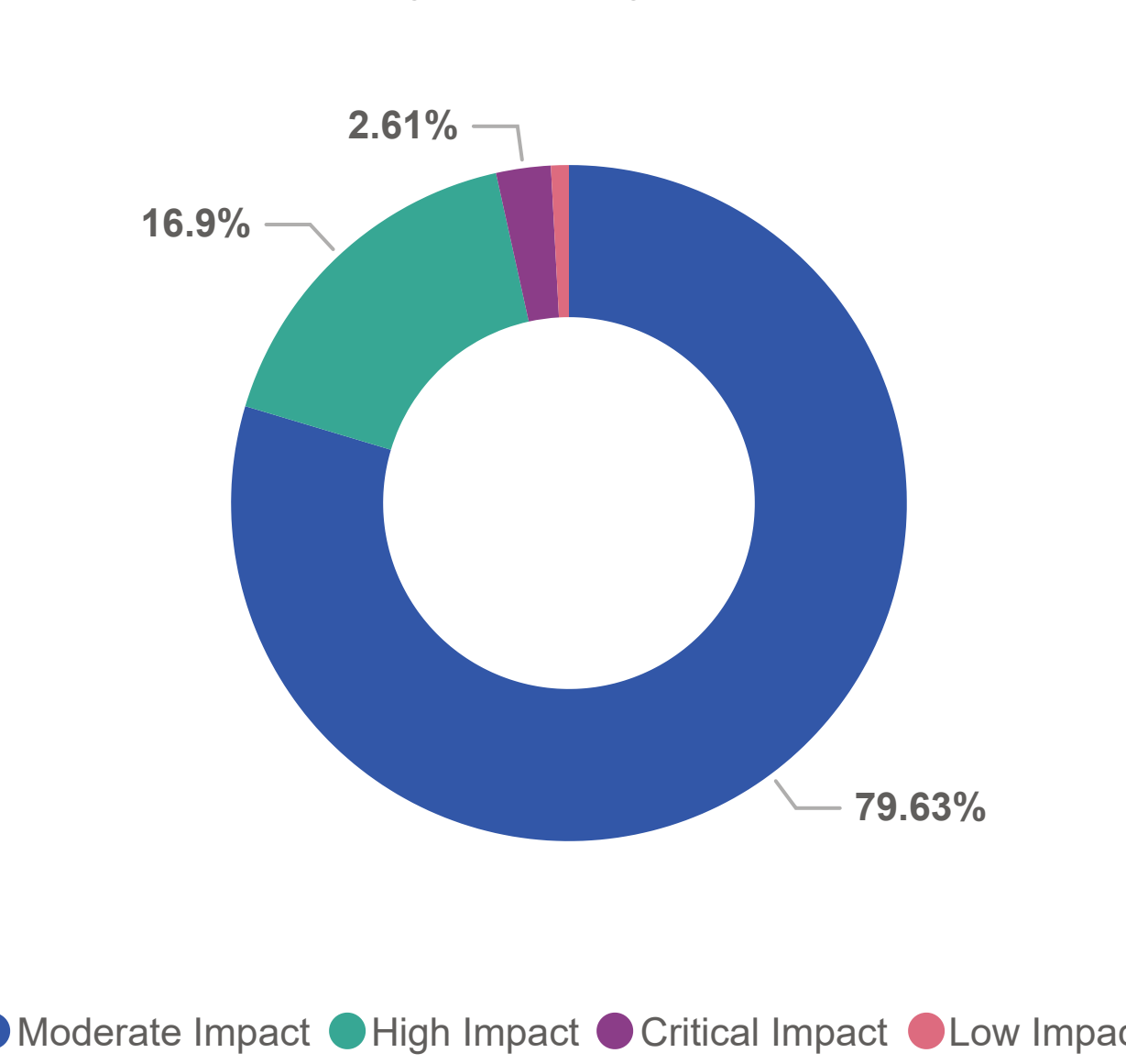
421.9

Avg Duration

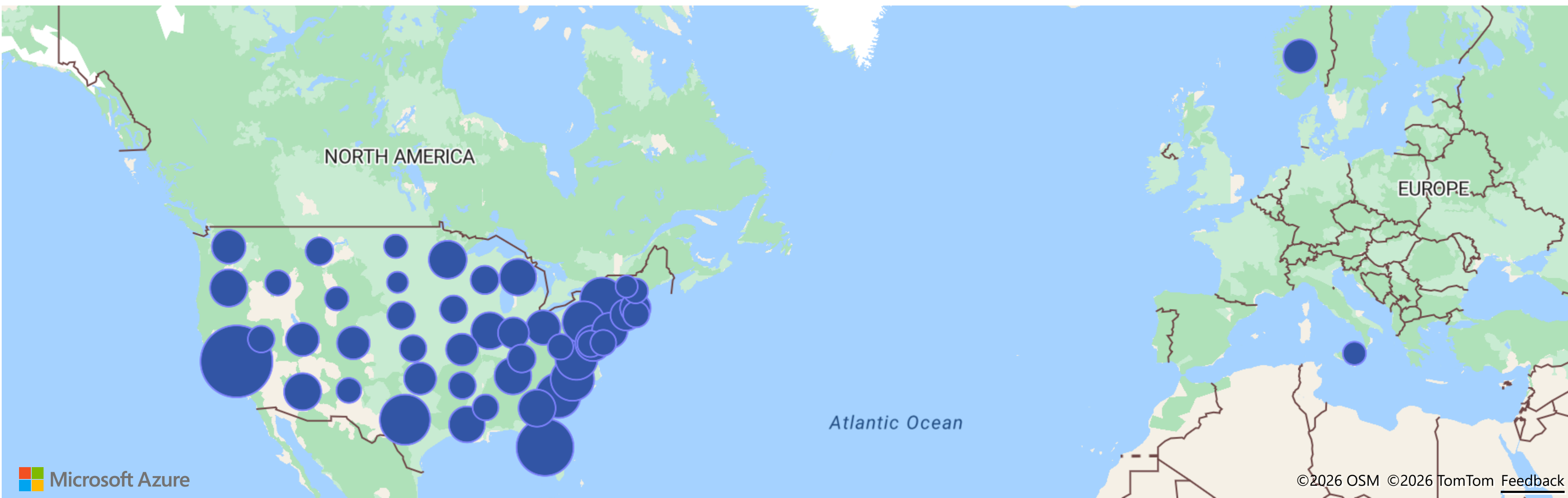
19.52

High Severity %

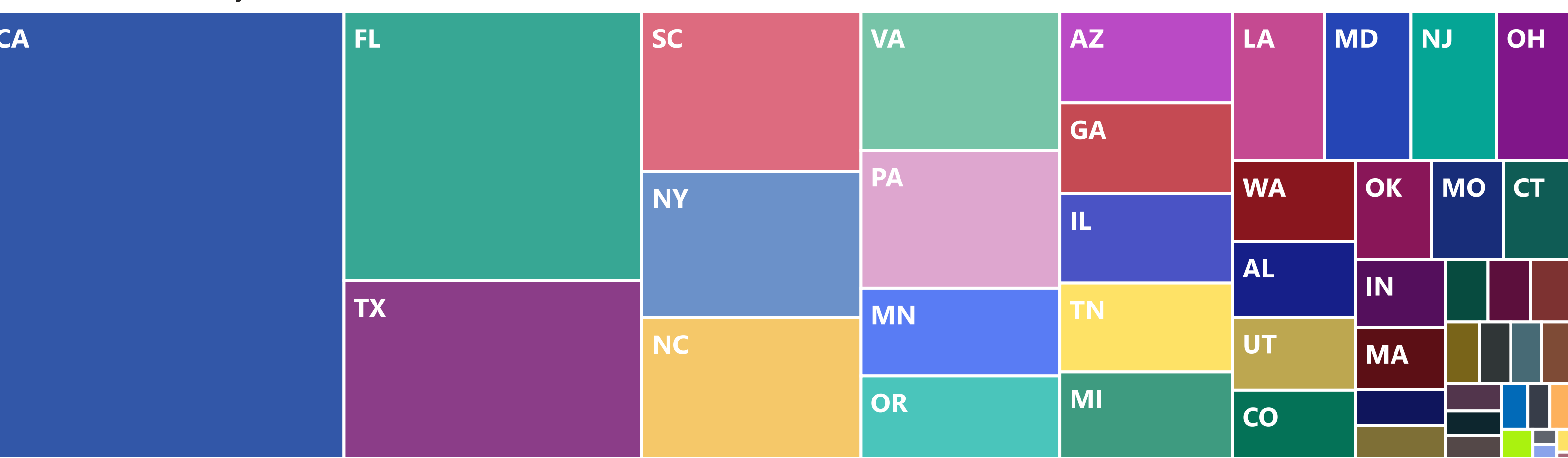
Total Accidents by Severity



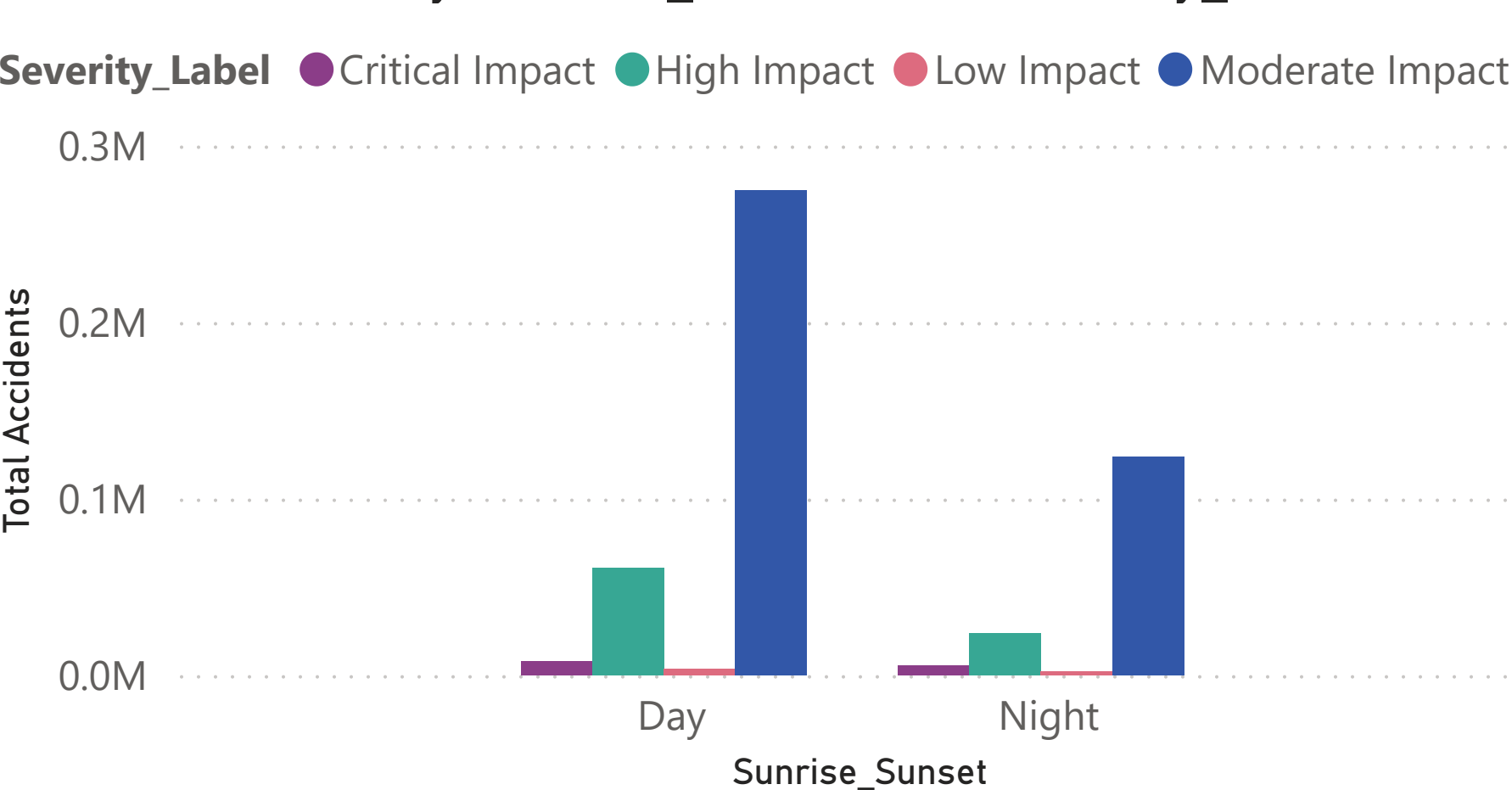
Total Accidents and Avg Severity by State



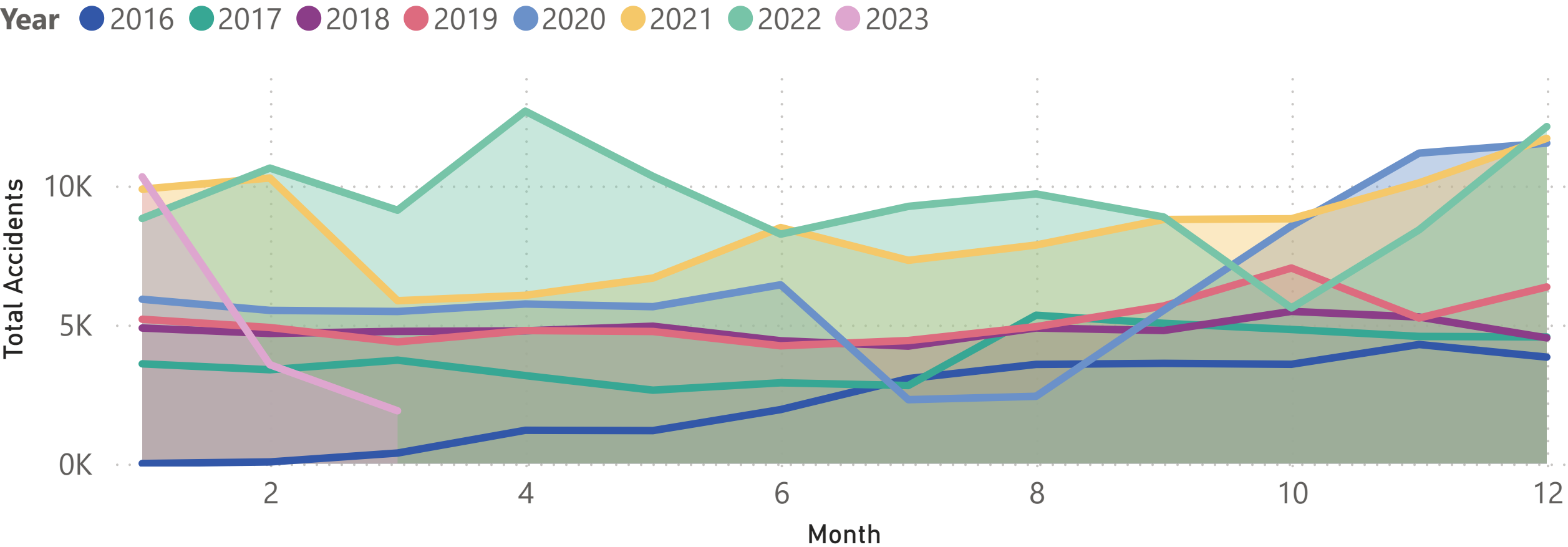
Total Accidents by State



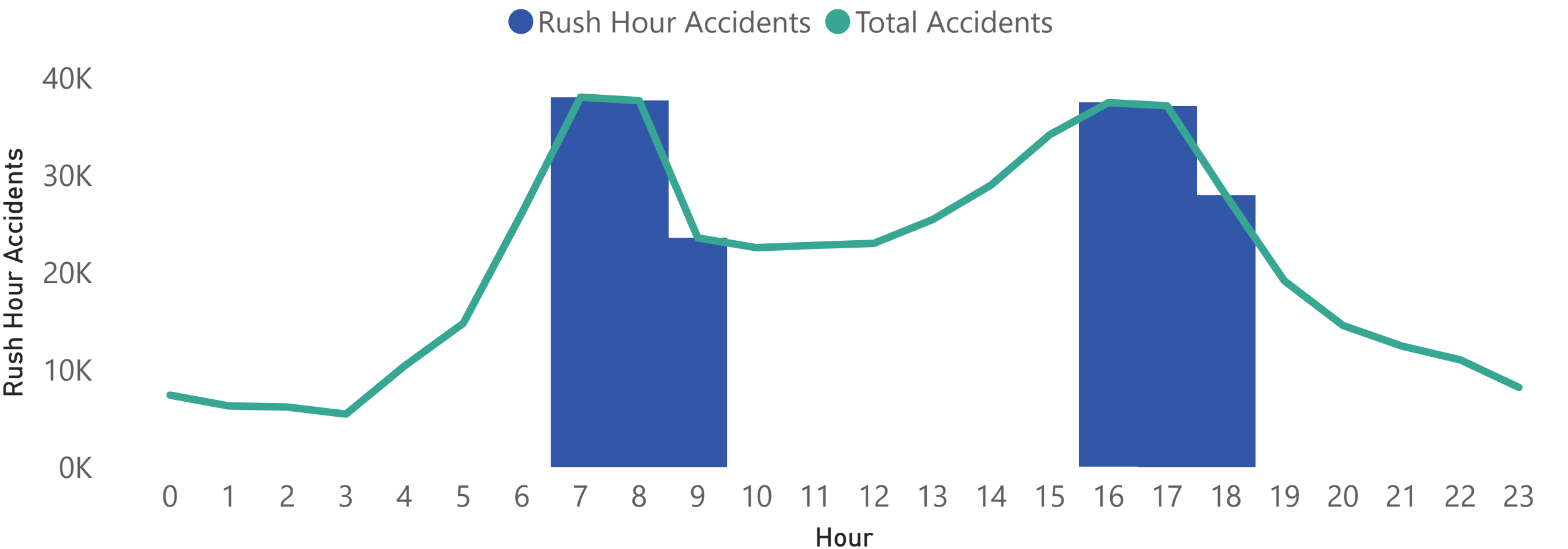
Total Accidents by Sunrise_Sunset and Severity_Label



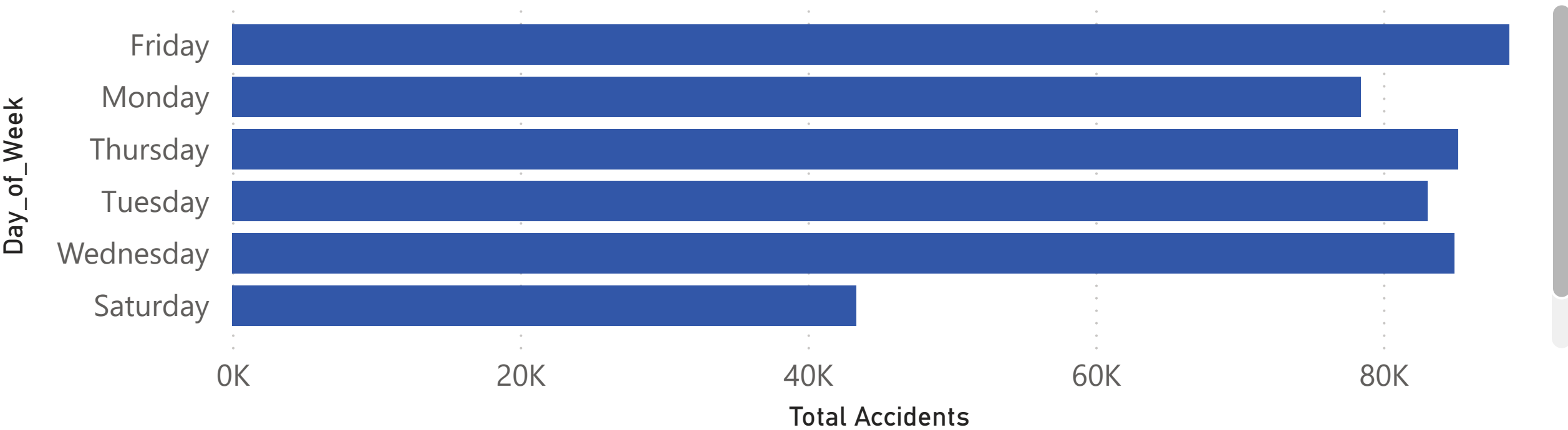
Total Accidents by Month and Year



Rush Hour Accidents and Total Accidents by Hour



Total Accidents by Day_of_Week



Year

All

Month

All

Severity_Label

All

State

All

202K

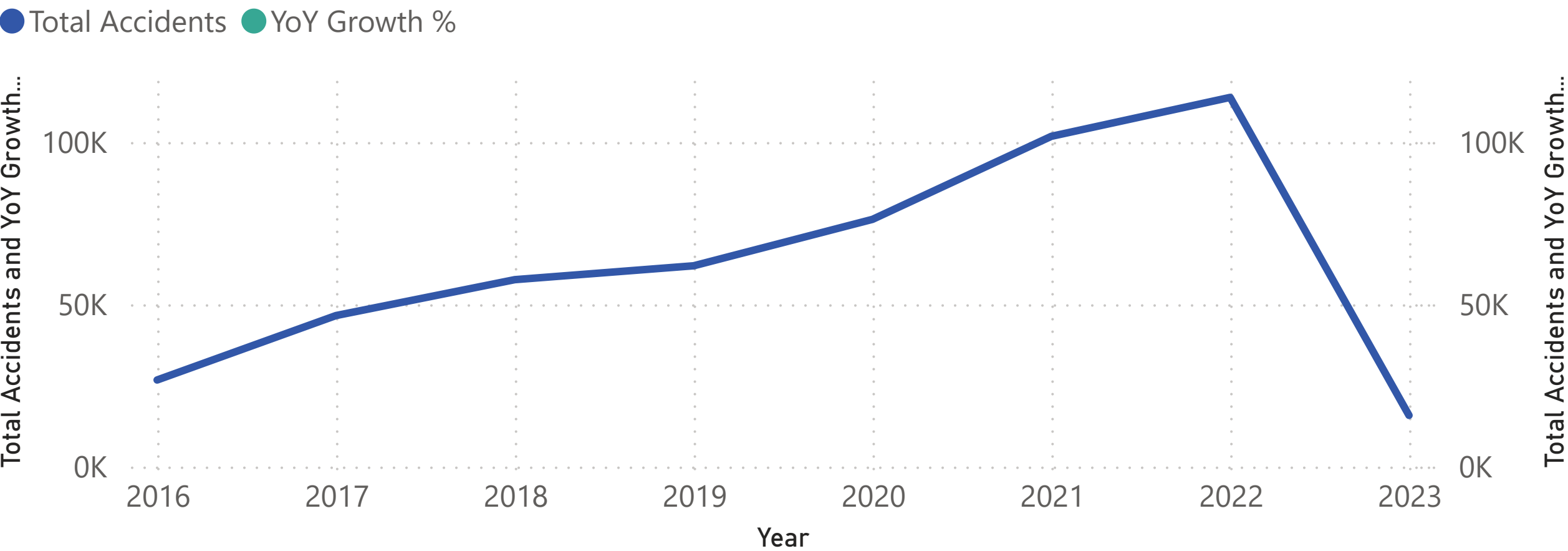
Rush Hour Accidents

80K

Weekend Accidents

Month	2016	2017	2018	2019	2020	2021	2022	2023	Total
1	2	3586	4874	5189	5909	9868	8813	10312	48553
2	58	3374	4679	4892	5508	10270	10626	3555	42962
3	378	3718	4754	4375	5469	5856	9110	1897	35557
4	1196	3164	4773	4783	5738	6050	12674		38378
5	1185	2639	4939	4749	5641	6671	10330		36154
6	1941	2904	4416	4234	6429	8487	8249		36660
7	3055	2806	4223	4424	2297	7309	9249		33363
8	3566	5334	4875	4923	2418	7858	9696		38670
9	3603	5043	4784	5667	5514	8777	8867		42255
10	3570	4822	5472	7032	8545	8803	5595		43839
11	4281	4571	5270	5233	11164	10095	8406		49020
12	3338	4553	4510	6351	11533	11606	13110		51560
Total	26663	46514	57578	61852	76155	101740	113734	15764	500000

Total Accidents and YoY Growth % by Year



Year

All

State

All

Severity_Label

All

Weather Type

All

26K

Extreme Weather Accidents

37K

Rain Accidents

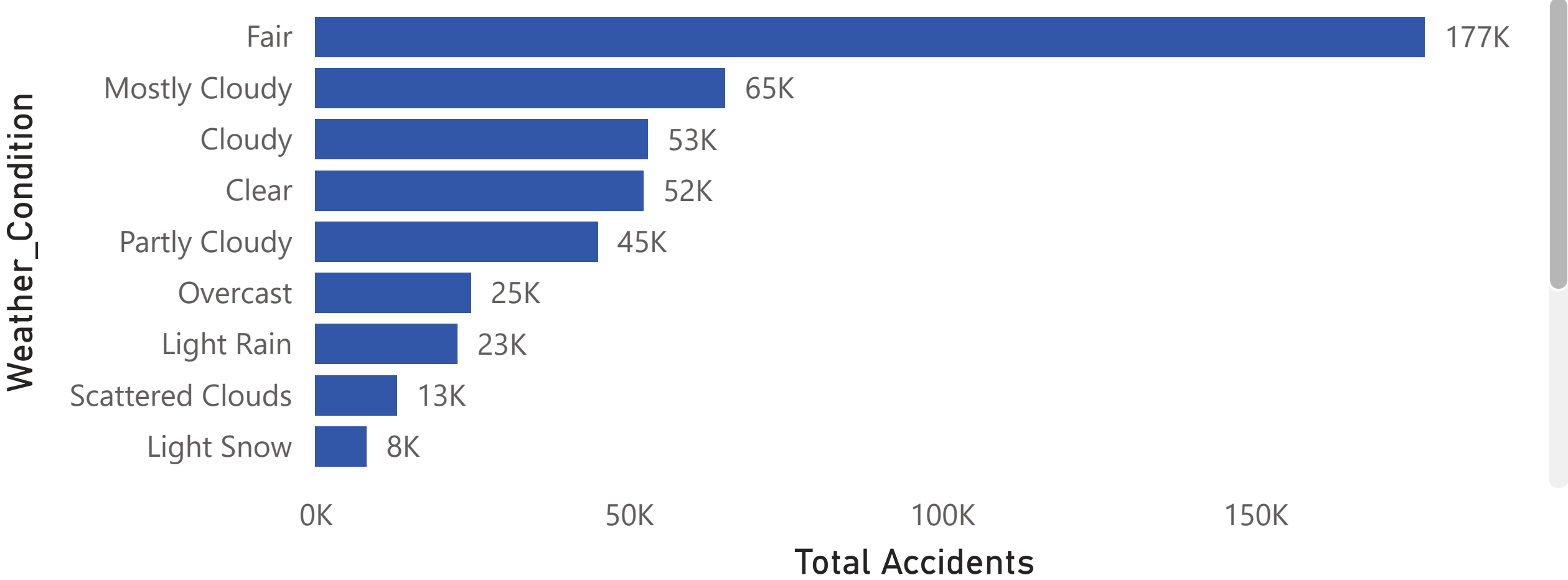
10K

Snow Accidents

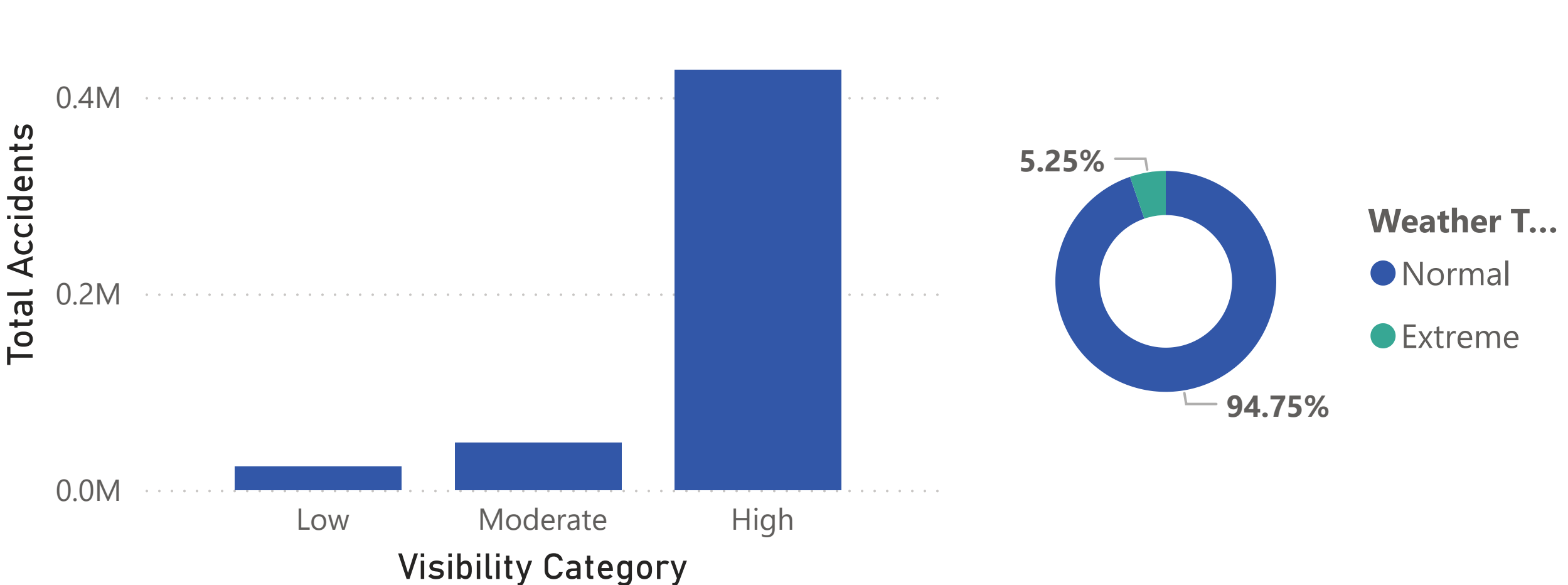
12K

Fog Accidents

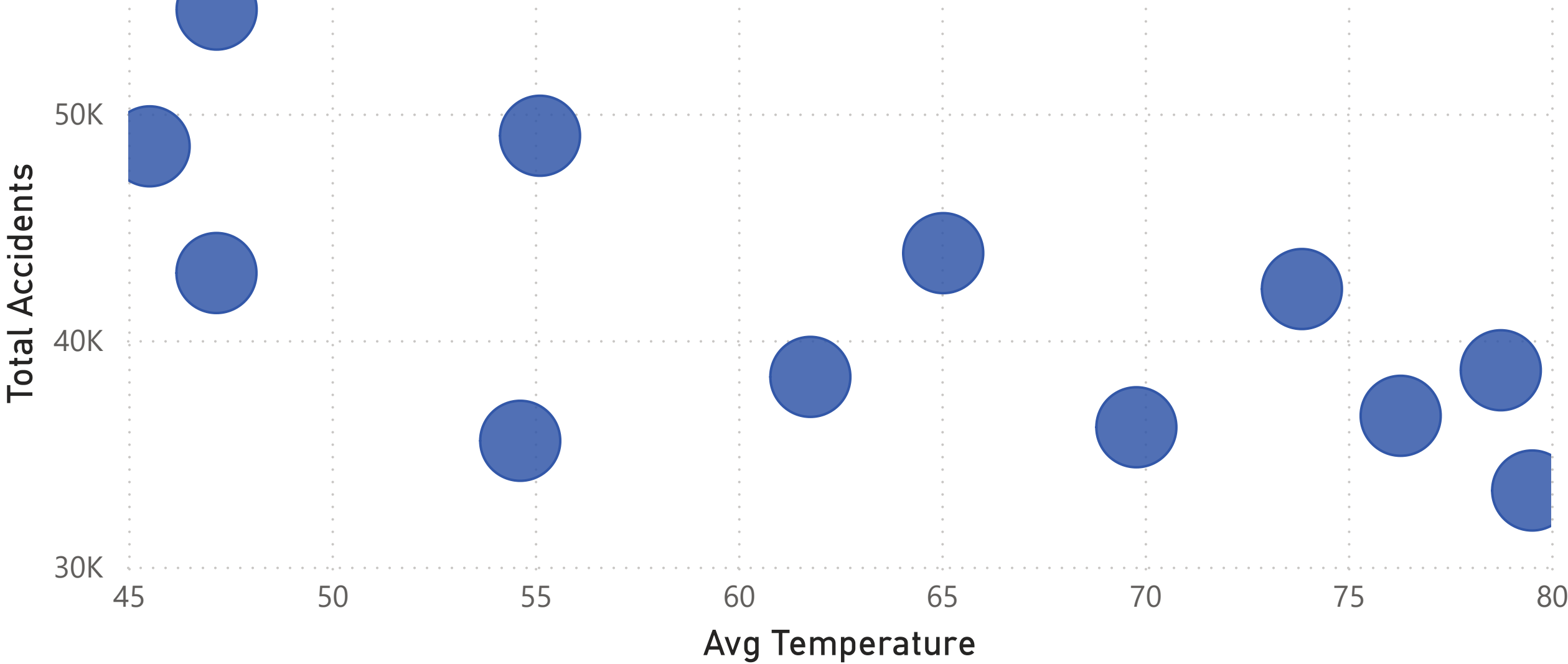
Top 15 Weather Conditions by Accidents



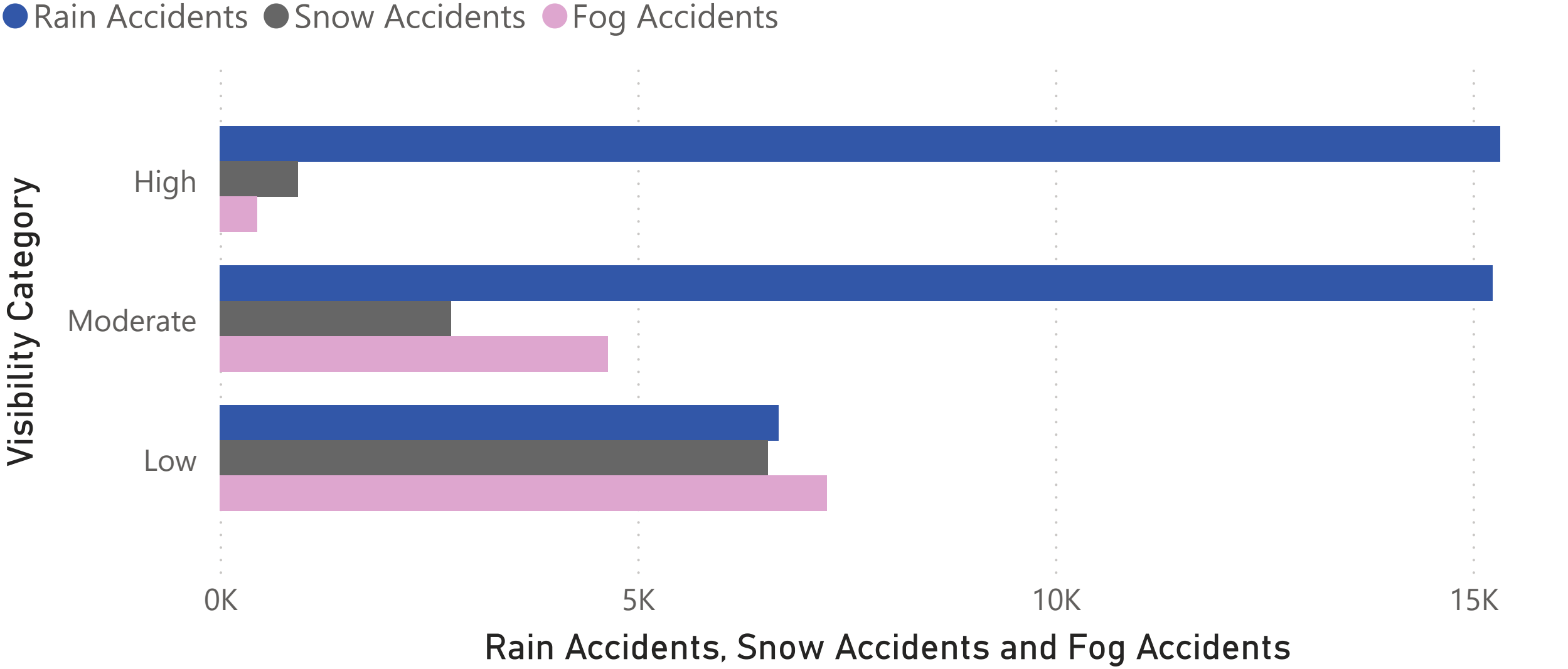
Total Accidents by Visibility Category



Avg Temperature, Total Accidents and Avg Severity by Month



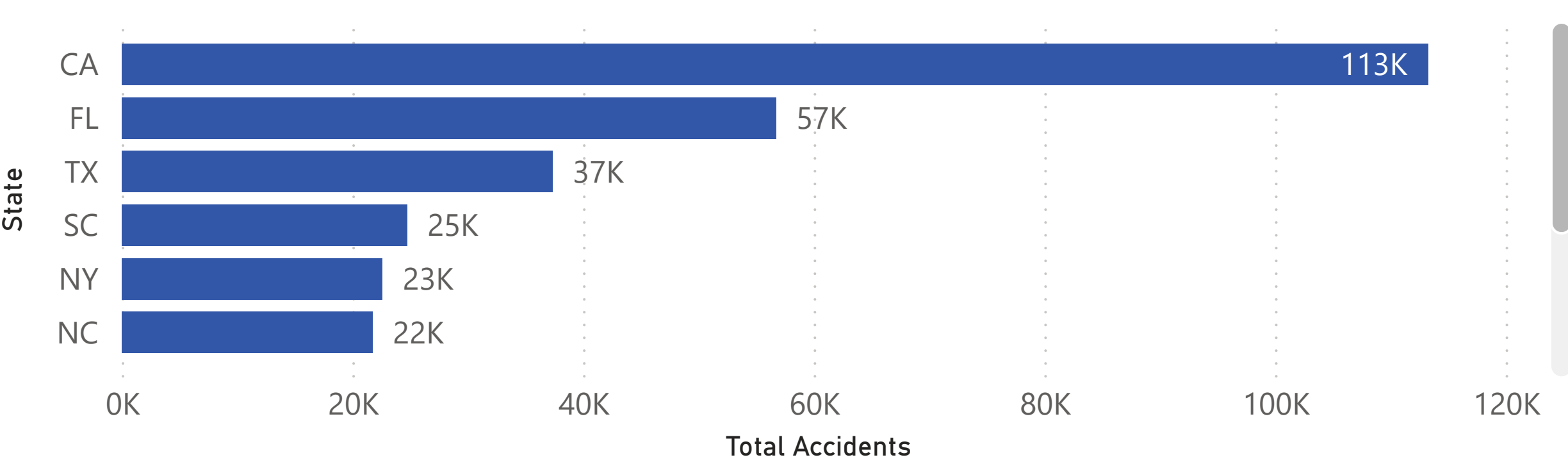
Rain Accidents, Snow Accidents and Fog Accidents by Visibility Category



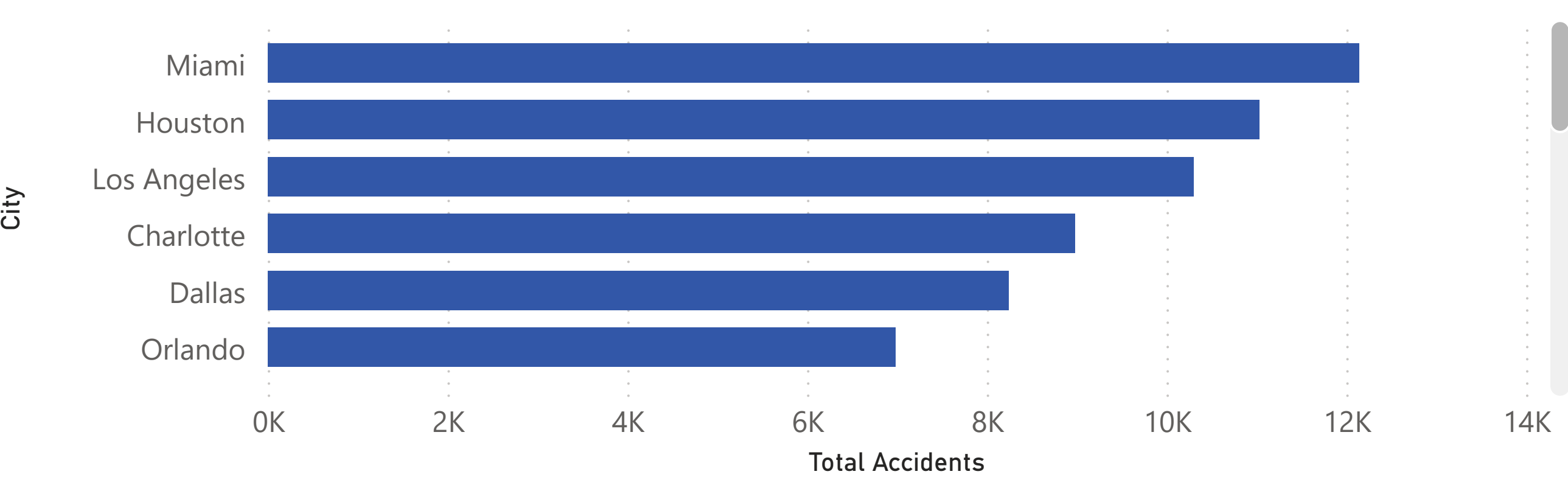
Total Accidents and Avg Severity by State



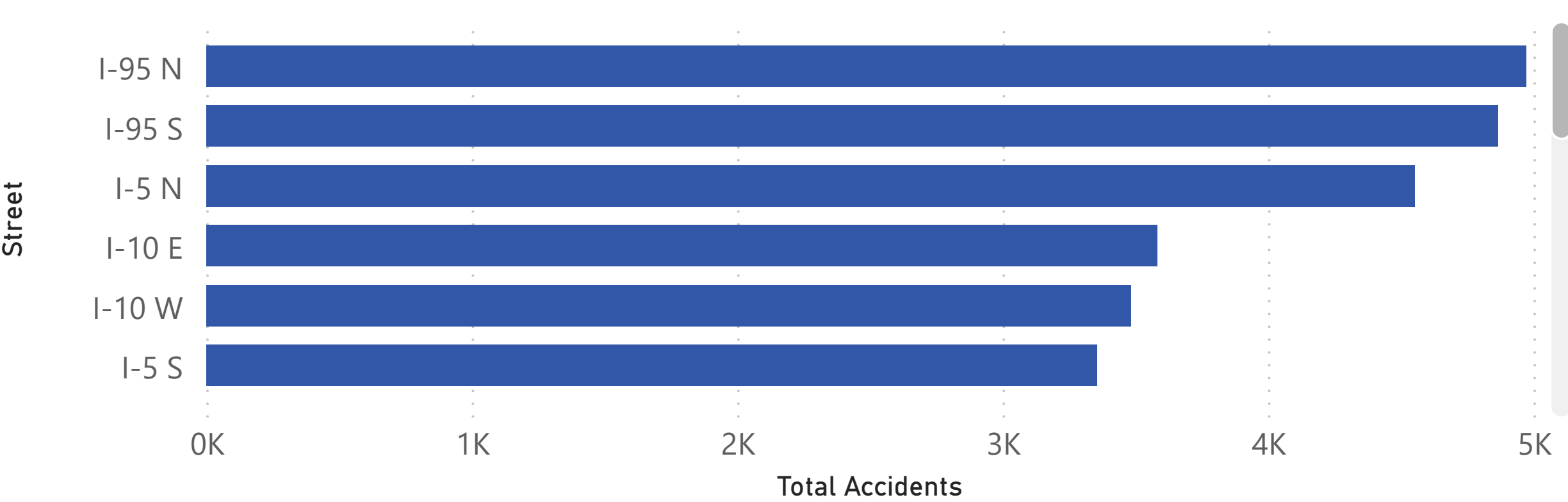
Total Accidents and Avg Severity by State



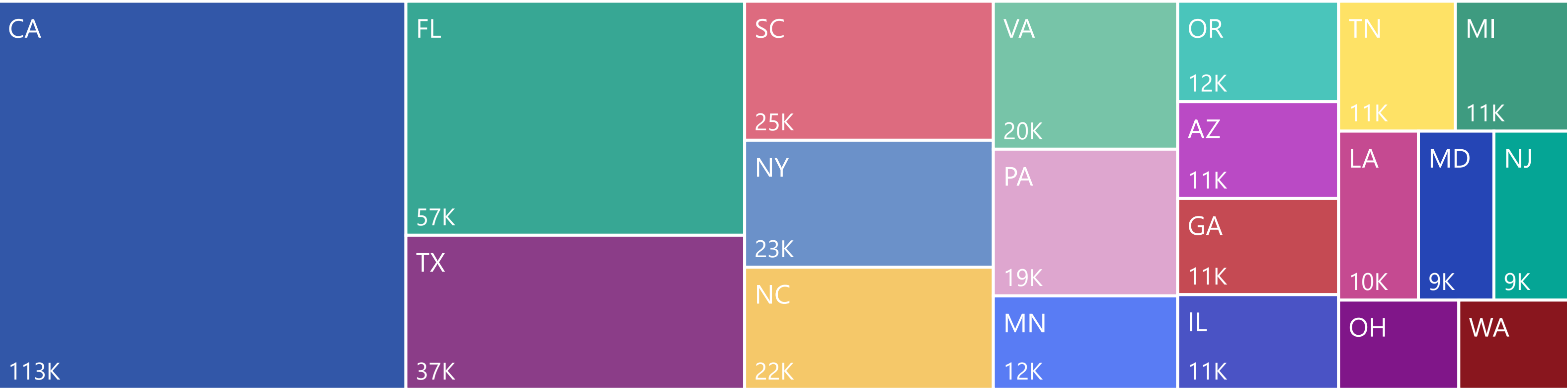
Top 20 High-Accident Cities



Total Accidents by Street



Accidents by State



State	Critical Impact	High Impact	Low Impact	Moderate Impact	Total
⊕ AL	113	1423	46	5003	6585
⊕ AR	194	37	1	1251	1483
⊕ AZ	348	1104	436	9262	11150
⊕ CA	866	17758	678	93972	113274
⊕ CO	458	1725	63	3678	5924
⊕ CT	286	1091	5	3220	4602
⊖ DE	42	50	7	1000	1099
Total	13064	84520	4274	398142	500000

Year

All

State

All

Severity_Label

All

Rush Hour Status

All

74K

Signal Accidents

37K

Junction Accidents

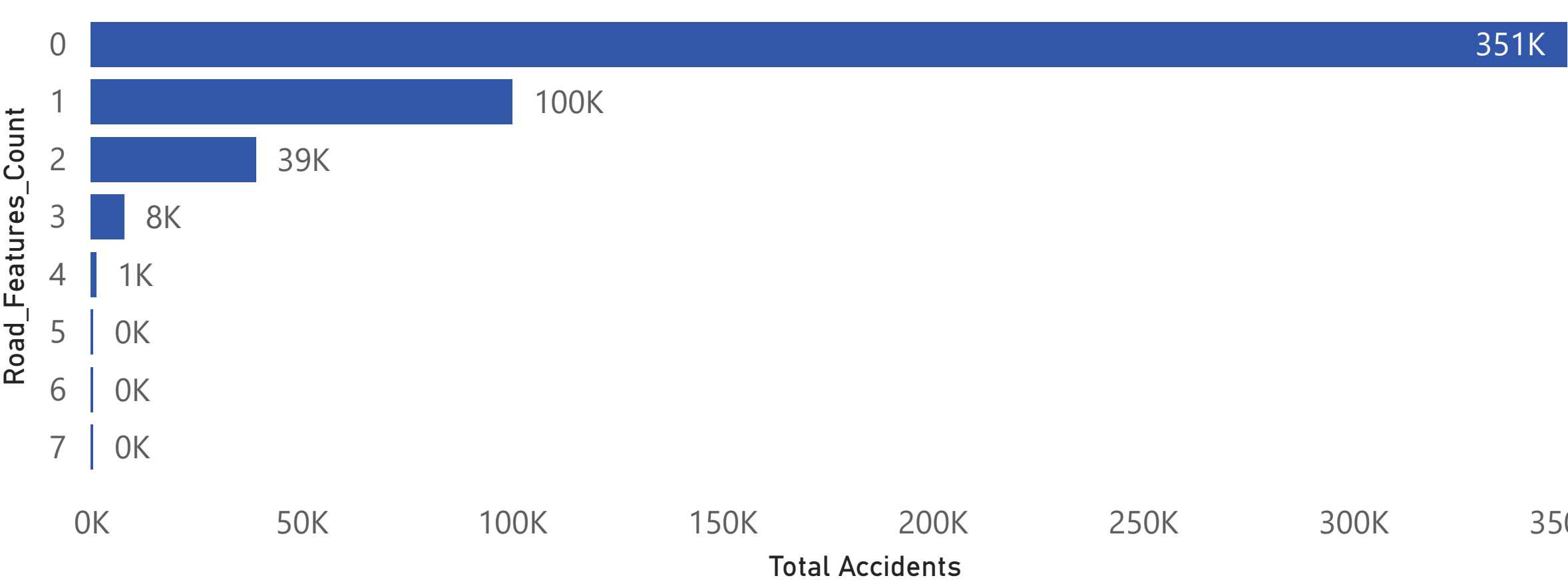
56K

Crossing Accidents

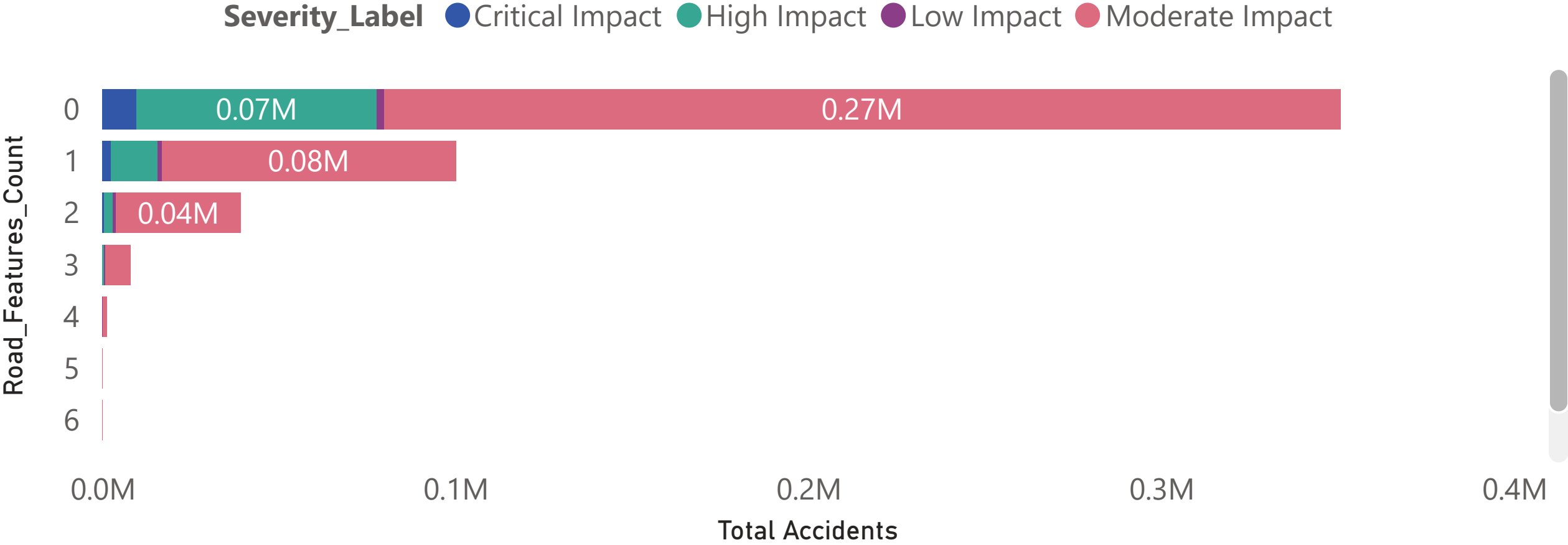
0.42

Avg Road Features

Accidents by Road Feature Presence

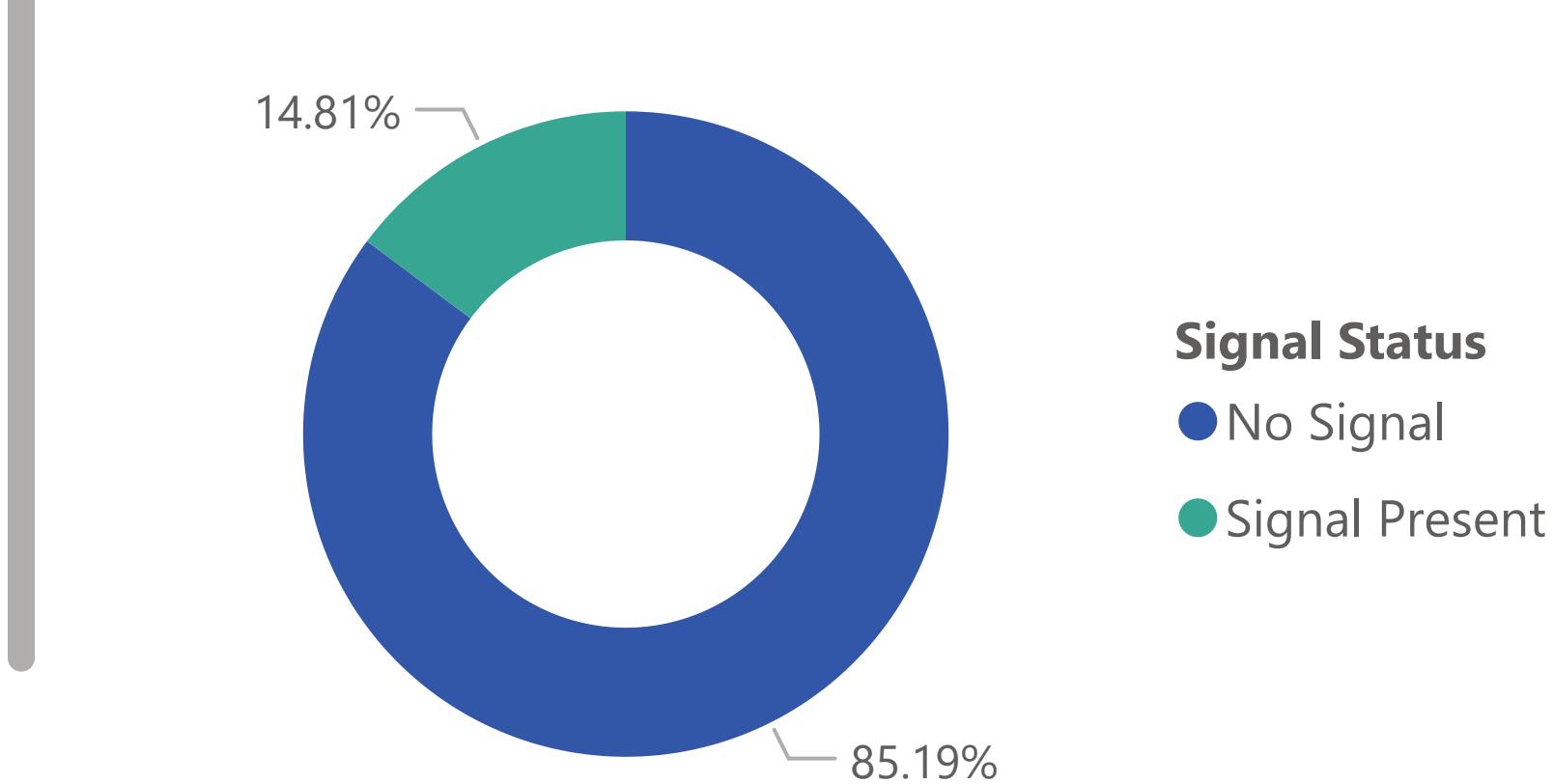


Total Accidents by Road_Features_Count and Severity_Label

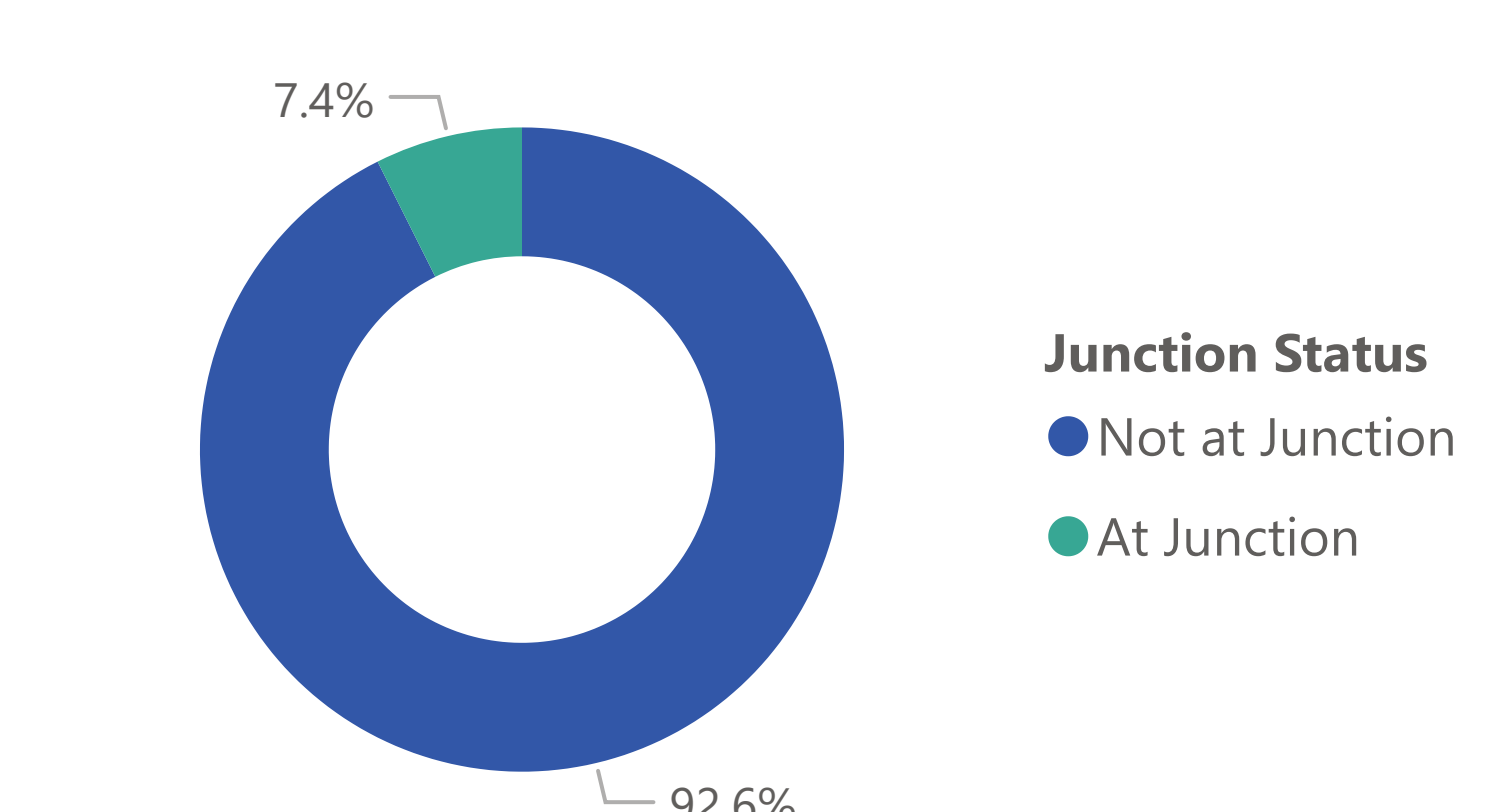


Road_Feature	Critical Impact	High Impact	Low Impact	Moderate Impact
Amenity	132	276	87	5696
Bump	6	20	2	183
Crossing	816	3185	1171	51081
Give_Way	82	314	40	1926
Junction	1324	8693	208	26782
No_Exit	37	112	25	1071
Railway	93	566	69	3592
Roundabout				13
Station	170	819	174	11835

Total Accidents by Signal Status



Total Accidents by Junction Status



Project Executive Summary



- **Architecture:** Medallion Pipeline (Bronze > Silver > Gold) using Python for ETL and Spark/SQL for data refinement.
- **Data Volume:** Processed and analyzed over **500,000 accident records**, ensuring data quality and integrity throughout the lifecycle.
- **Modeling:** Developed a Multi-class Classification model using **LightGBM**, achieving an optimized **Accuracy of 69.83%**.
- **Best Model:** CatBoost (Binary Classification) — ROC-AUC: 0.746, Threshold: 0.5034.

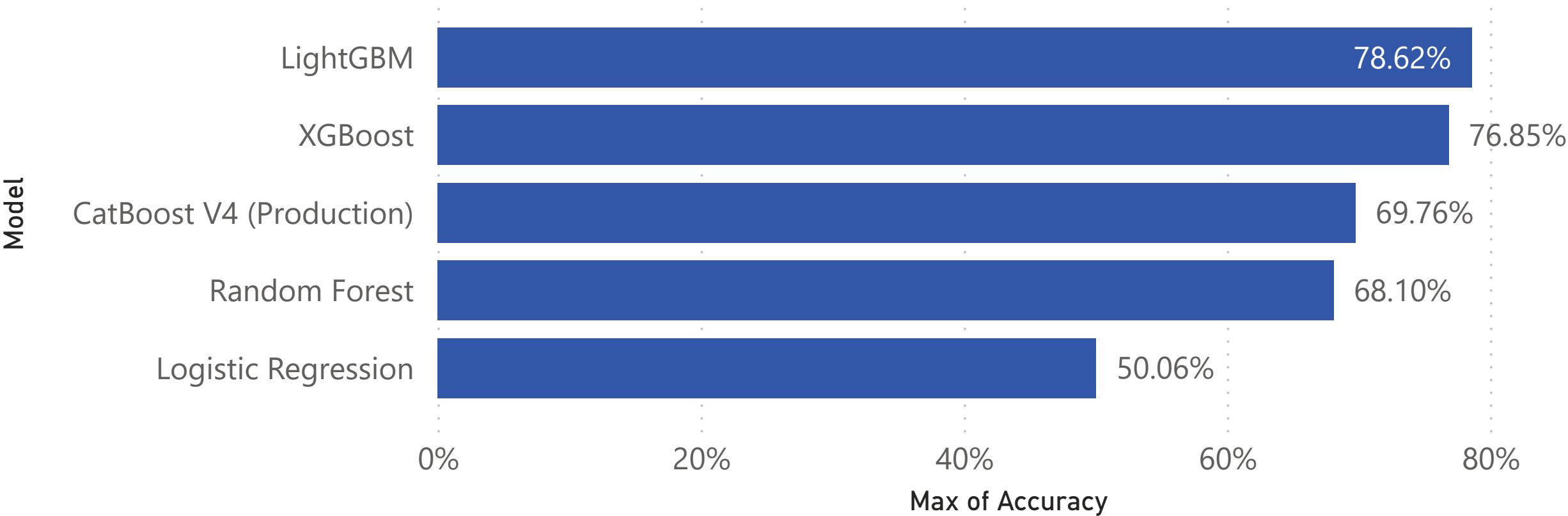
78.62%

Accuracy

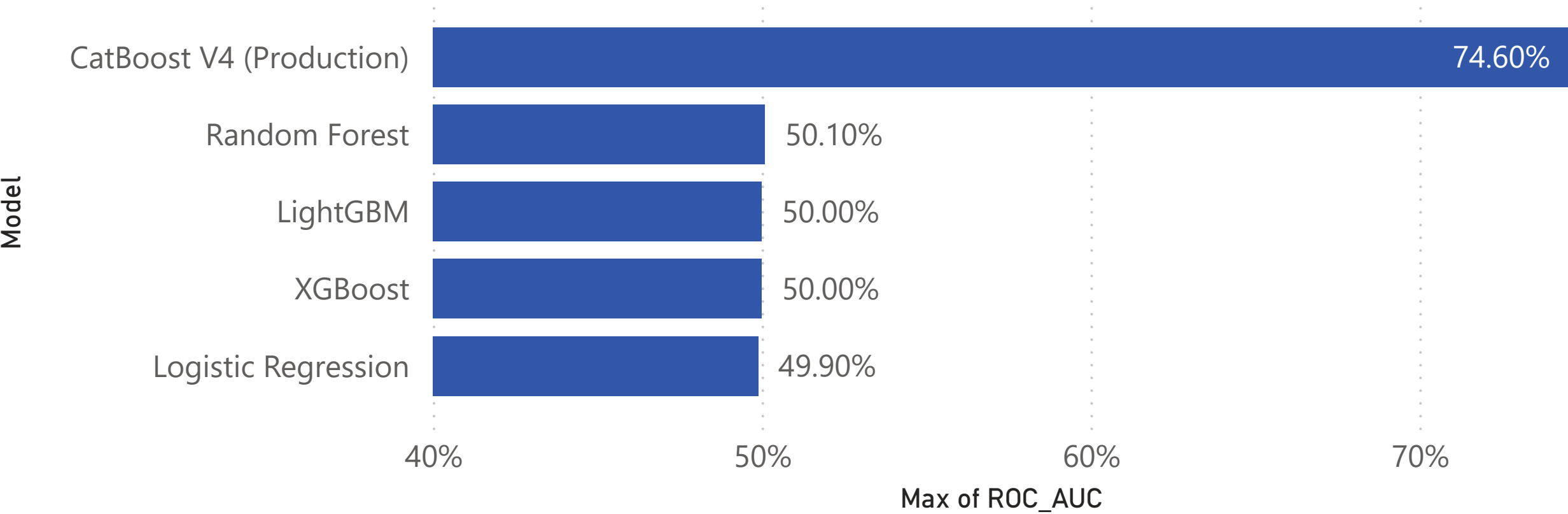
30

Features Count

Accuracy by Model

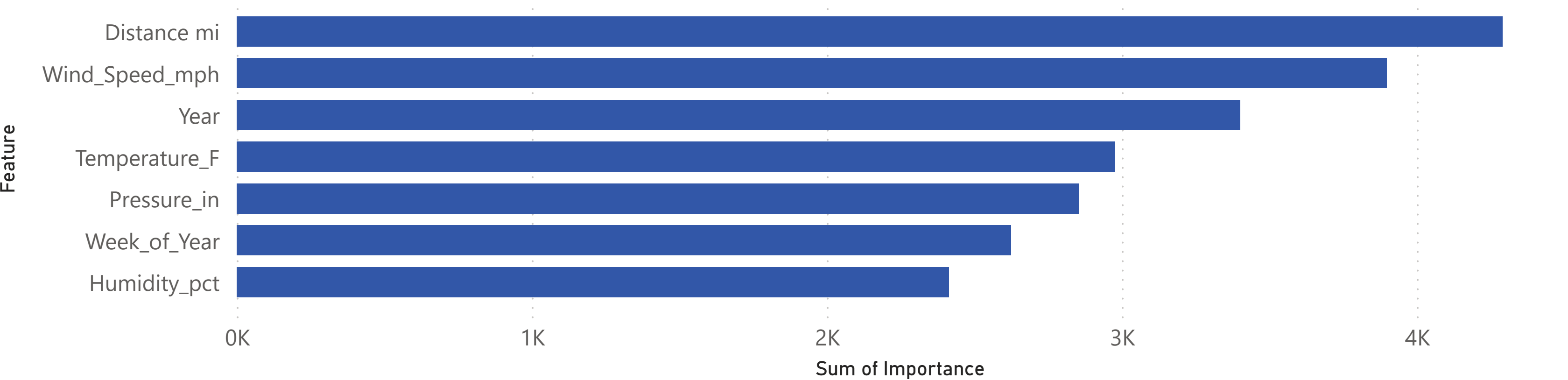


ROC_AUC by Model



Model	Task	Accuracy	ROC_AUC
CatBoost V4 (Production)	Binary	69.76%	74.60%
Random Forest	Binary	68.10%	50.10%
LightGBM	Binary	78.62%	50.00%
XGBoost	Binary	76.85%	50.00%
Logistic Regression	Binary	50.06%	49.90%
LightGBM	Multi	69.83%	
Logistic Regression	Multi	43.94%	
Random Forest	Multi	66.41%	
XGBoost	Multi	68.34%	

Sum of Importance by Feature



Forecast Data

Historical Data



497K

Total Historical Accidents

124K

Total Predicted Next 12 Months

14K

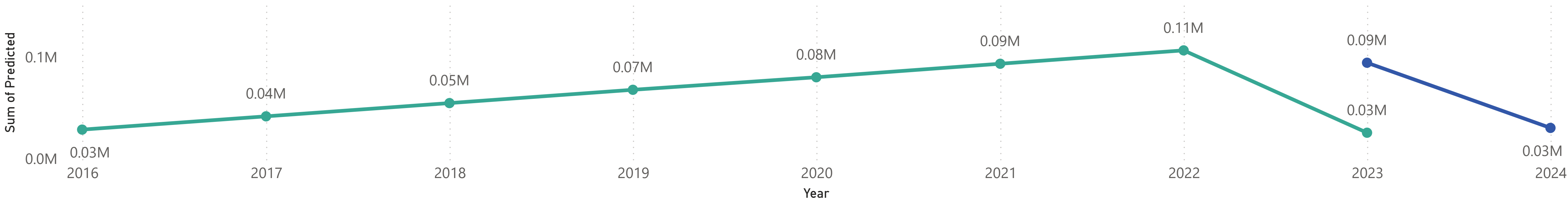
Peak Month Prediction

7K

Lowest Month Prediction

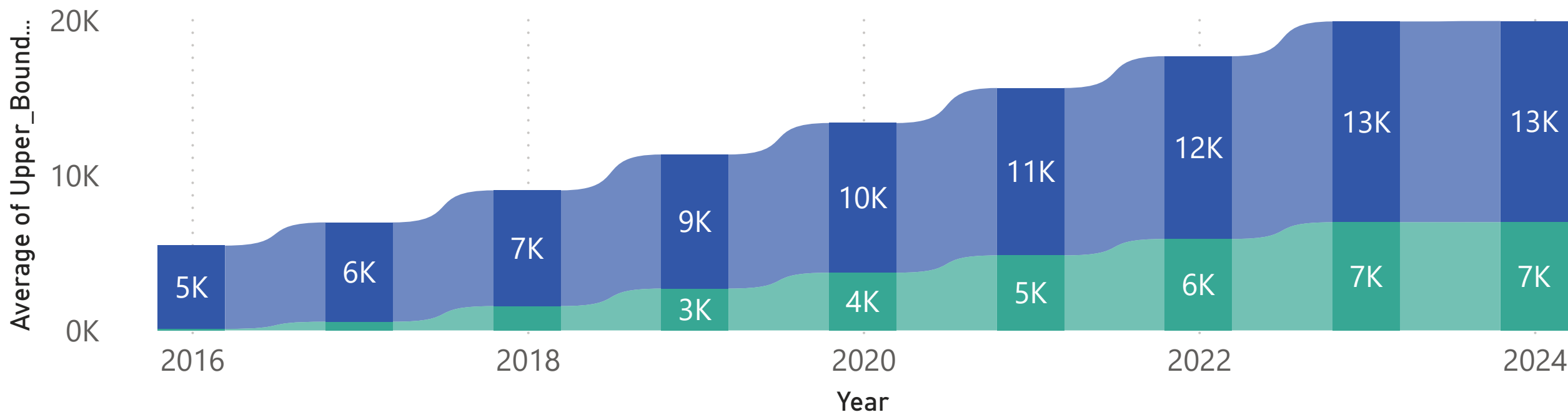
US Accidents — 12 Month Forecast (Prophet Model)

Forecast_Status ● Forecast Data ● Historical Data

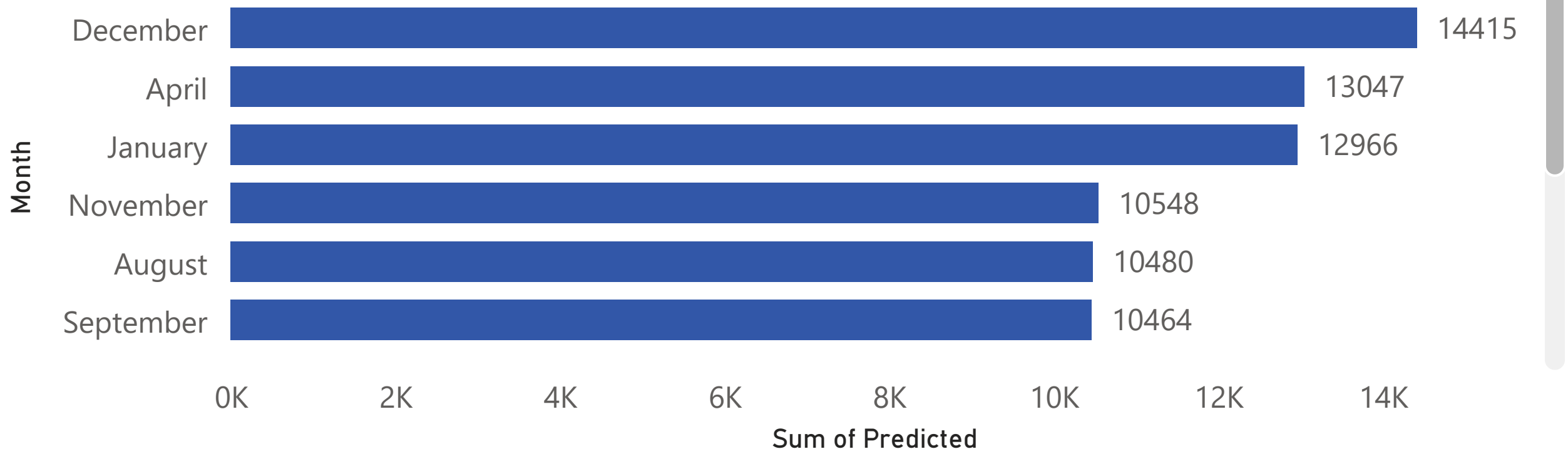


Forecast Confidence Interval (95%)

● Average of Upper_Bound ● Average of Lower_Bound



Predicted Accidents per Month



"Forecast Insights"

- December 2023 predicted as highest risk month: 14,415 accidents
- June 2023 predicted as lowest risk month: 6,731 accidents
- Winter months (Dec-Jan) show 40% more accidents than summer
- Model trained on 87 months of historical data (2016-2023)
- 95% confidence interval ensures reliable prediction range

363

Total Predicted High Risk

637

Total Predicted Low Risk

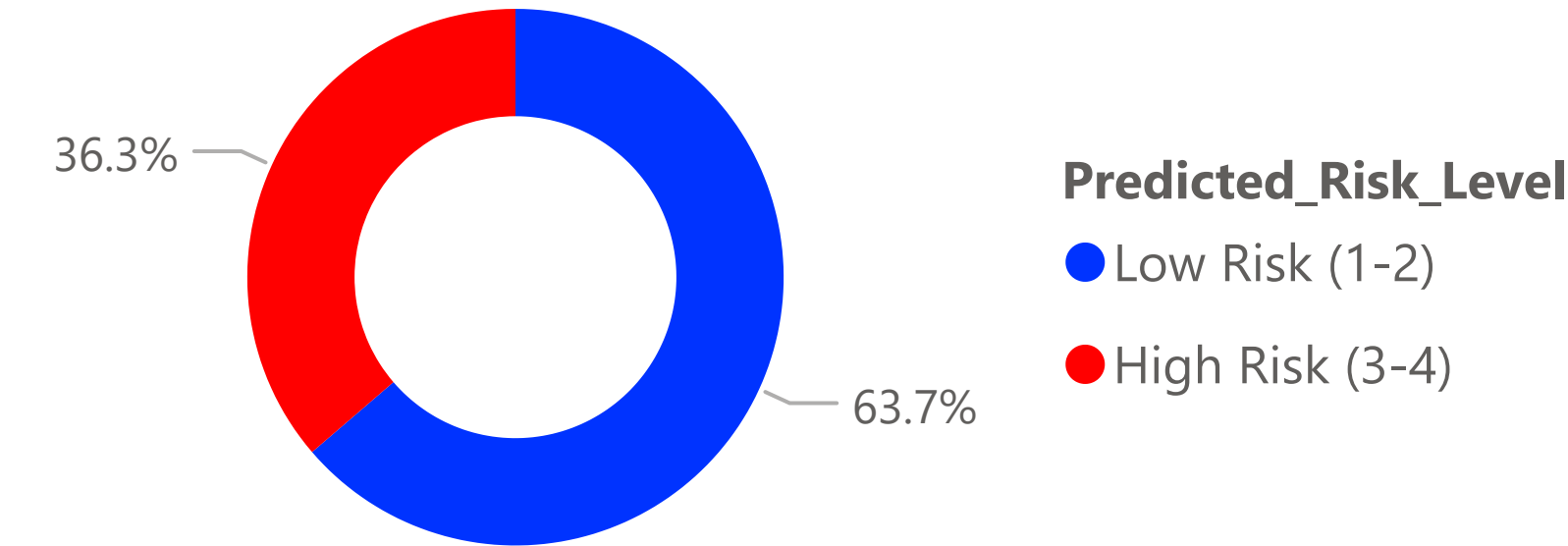
0.44

Avg Risk Score

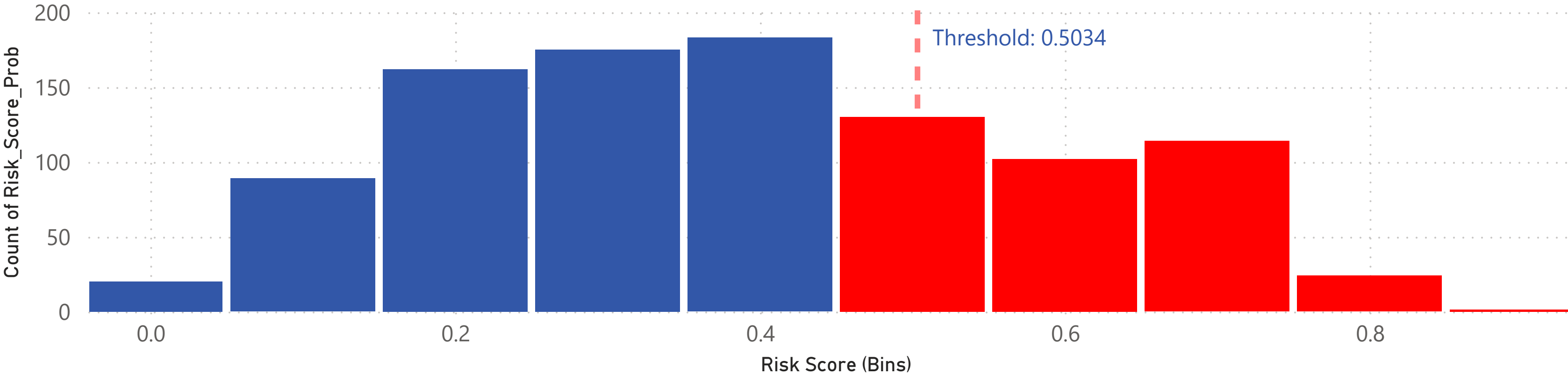
36.30

High Risk %

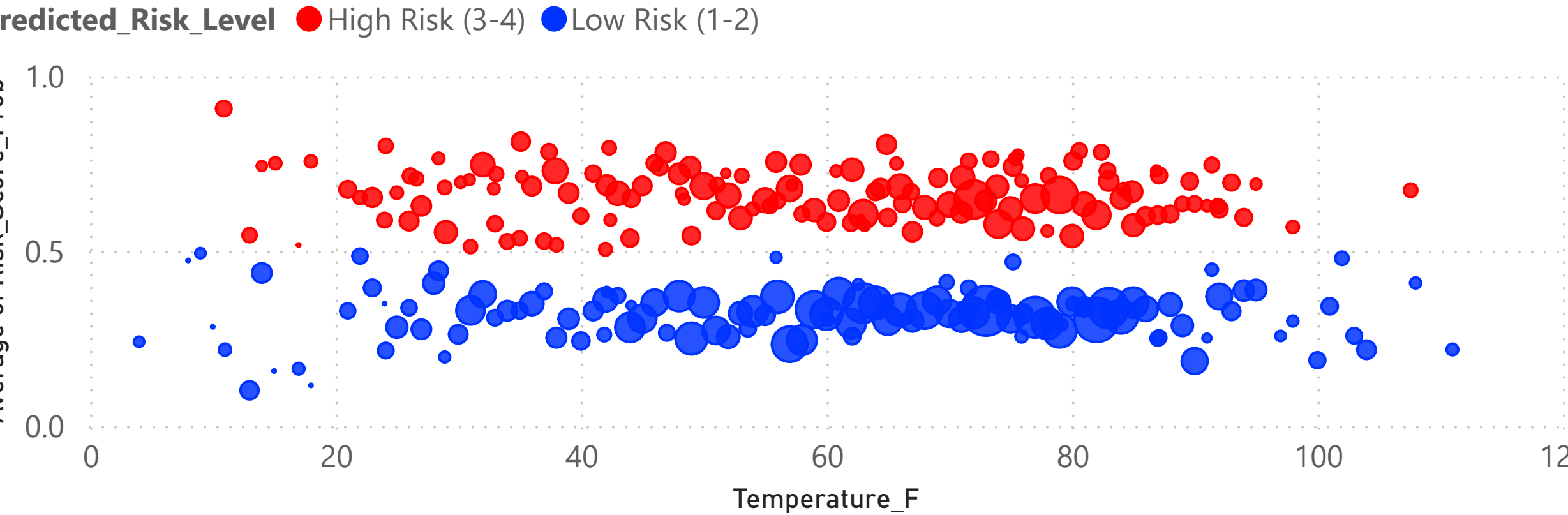
Count of Predicted_Risk_Level by Predicted_Risk_Level



Count of Risk_Score_Prob by Risk Score (Bins)



Average of Risk_Score_Prob and Sum of Wind_Speed_mph by Predicted_Risk_Level and Temperature_F



Year	Weather_Condition_Encoded	Predicted_Risk_Level	Average of Risk_Score_Prob
2016	42	High Risk (3-4)	0.79
2016	62	High Risk (3-4)	0.77
2018	42	High Risk (3-4)	0.76
2017	77	High Risk (3-4)	0.76
2016	77	High Risk (3-4)	0.76
2017	49	High Risk (3-4)	0.76
2016	65	High Risk (3-4)	0.76
2019	4	High Risk (3-4)	0.75
2019	65	High Risk (3-4)	0.75
2018	77	High Risk (3-4)	0.75
Total			0.44